
Gentrification and Suburbanization

LSE GY457 - AT 2025

Topic 9

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Course overview

- Block 1

- Intro to Urban and Regional Economics and course overview
- Topic 1: Regional and urban concentration forces
- Topic 2: The empirics of agglomeration
- Topic 3: The costs and benefits of agglomeration

- Block 2

- Topic 4&5: Monocentric city model
- Topic 6: Firm location choice

- Block 3

- Topic 7: The vertical dimension of cities
- Topic 8: Neighborhoods and segregation
- **Topic 9: Gentrification and suburbanization**
- Topic 10: Measuring cities with new technologies

Recap from last time

- The choice of house embeds choice of neighborhood too
 - Nhbd demographics influence housing demand and prices
- Segregation in the US peaked in 1970s, declining since then but strong persistence
 - Collective exclusion contributed to early segregation
 - White flight prominent throughout (to suburbs after 1940s)
 - Urban forms can contribute to segregation in cities
- Segregated cities perform poorly on important socio-economic outcomes
 - Poverty, crime, upward mobility, educational attainment,...
- Living in large enclaves can be beneficial at first, but may hinder assimilation and economic outcomes

Today: Gentrification and Suburbanization

- Determinants of long-run trends
 - Relative transportation costs by income groups
 - Lifecycle of buildings
 - Amenities
 - Endogenous gentrification
- Welfare dimension
 - Winners and losers
 - Role of mobility costs
- Consequences of gentrification

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Suburbanization

- Decentralization in the 20th century
 - Population movements from central cities to suburban areas
- Monocentric city model offers two explanations:
 1. Increase in income
 - If commuting costs do not depend on income
 2. Decrease in transportation costs due to better transport technology
- Suburbanization also associated with white flight

City centers used to be rich: Milwaukee 1860

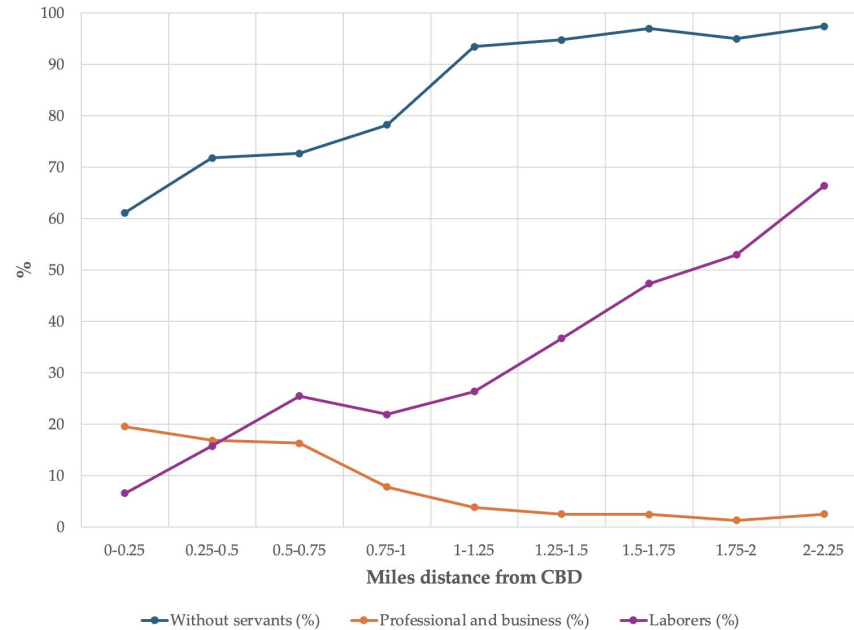


Figure based on data reported in Table 1 of LeRoy and Sonstelie (1981)

White flight

TABLE 4
DISTRIBUTION OF PERCENTAGE BLACK IN CENSUS TRACTS

	1940: City	1960		1990	
		City	Suburbs	City	Suburbs
Number of tracts	6,113	13,310	9,378	16,664	27,183
Percentage of tracts with black share:					
Exactly zero	21.2	19.6	22.3	7.3	14.7
0–1	39.1	36.2	48.0	10.2	25.0
1–5	15.6	12.4	13.3	23.7	30.0
5–15	8.3	8.8	8.2	18.8	15.3
15–25	3.6	4.2	2.9	7.4	5.3
25–50	4.2	5.4	2.9	9.4	4.9
50–75	3.2	4.6	1.3	7.0	2.4
75–90	1.9	3.4	.6	5.0	1.1
90–98	1.8	3.1	.3	6.5	.9
98–100	1.0	2.3	.2	4.8	.4

NOTE.—The sample is census tracts with at least some population.

Source: Cutler et al. (1999)

Why did middle class households moved to suburbs over the 20th century?

Gentrification in US cities 1990-2010

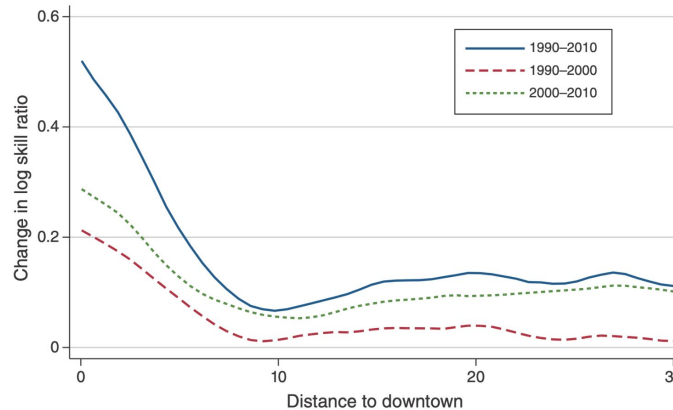


FIGURE 2. CHANGE IN SKILL RATIO BY DISTANCE TO DOWNTOWN

Notes: The graph shows a nonparametric plot between the change in log skill ratio and census tracts' distances to downtowns in the top 25 most populous MSAs (defined by population ranking in 1990). The solid line represents the change in log skill ratios from 1990 to 2010. The long-dashed line represents the change in log skill ratios from 1990 to 2000, and the short-dashed line represents the change from 2000 to 2010.

Source: The source of the data is the decennial census and ACS provided by NHGIS.

Source: Su (2022)

High skilled are moving back to city centers since the 1990s

Core vs. suburbs in US cities 1950-2010

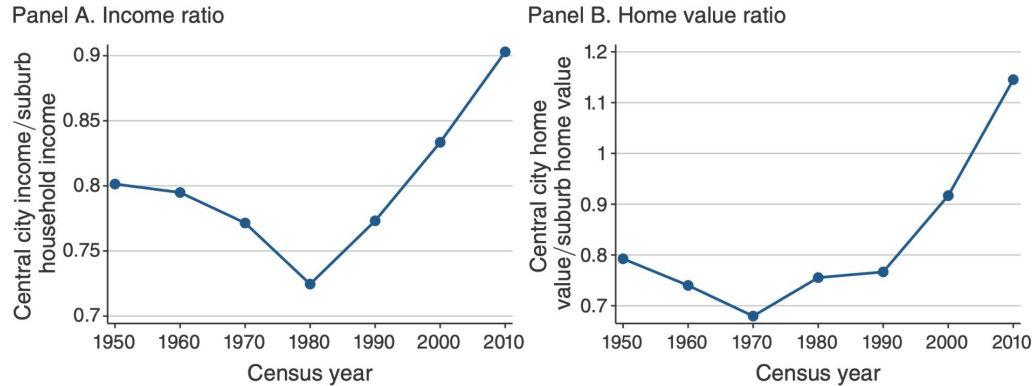


FIGURE 1. INCOME AND HOME VALUE RATIO BETWEEN CENTRAL CITY AND SUBURBAN NEIGHBORHOODS

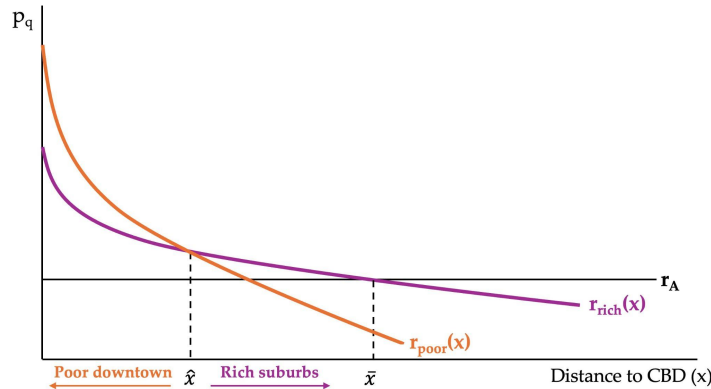
Notes: Central cities in these figures are census tracts that are located within five miles of the downtown in the respective MSAs defined in Holian and Kahn (2015). The values plotted are the mean income and home value of the census tracts located in the central cities and the mean income and home value of noncentral city census tracts in the top 25 MSAs (defined by population ranking in 1990).

Sources: The source of the data is the decennial census and ACS provided by NHGIS.

Source: Su (2022)

Gentrification associated to rising income and house value

Income segregation



- Being able to predict life cycle of cities can be very profitable
- Monocentric city model (under some assumptions) can explain why rich live in suburbs and poor in centers
 - Static prediction
- Is there one theory that explains why rich move from city to suburbs and back while poor do the opposite?

Transportation cost cycle

- LeRoy and Sonstlie (1981) extend the monocentric city model framework *to include different transit modes*
- New modes are usually faster but also more expensive
 - Adopted first by the rich (they value time more)
 - Adopted later by the poorer (costs of new technologies tend to decrease + rising incomes over time)
 - As happened for trains, subways, cars... drones in the future?
- LeRoy and Sonstlie (1981) focus is on US cars adoption
 - Affordable to the rich already at beginning of 20th century
 - Affordable to almost everyone at the end of 20th century

Transportation cost cycle

- Assumptions:
 - 1 2 transport modes: (i) fast and expensive vs. (ii) cheap and slow
 - 2 Income elasticity of housing demand lower than marginal commuting costs (*intuition?*)
- Implications:
- Rich live in the center if rich and poor use same mode
 - i.e. if fast mode is unavailable/unaffordable to both types
 - or if fast mode is affordable to both
 - Standard scenario (workers can use only one transit mode)
- Rich live in suburbs if fast mode available to rich only
 - Novel scenario

Transportation cost cycle

- Can identify four periods depending on commuting modes by income (LeRoy and Sonstlie, 1981)
 1. *Paradise* (rich live in center)
 - Fast mode so expensive that all commuters use slow mode
 2. *Paradise lost* (rich move to suburbs)
 - Fast mode available to rich, unaffordable to poor
 - Downtown deteriorates
 3. *(Re)gentrification* (poor to suburbs, rich back to centers)
 - Fast mode available to rich and poor
 4. *Paradise regained* (poor live in suburbs)
 - All poor commute using fast mode
 - Downtown restored, but suburbs deteriorates

Car ownership over time

TABLE 5
Percentage of Households Owning Cars, by Income Quintile

Year	Quintile				
	Lowest	Second	Third	Fourth	Highest
1952	26	44	64	79	89
1957	33	62	82	89	95
1962	32	65	76	91	94
1969	45	79	93	95	97
1977	61	89	95	97	97

Sources. 1962 Survey of Consumer Finances [12], 1970 Survey of Consumer Finances [13], Federal Reserve Board, 1977 [4].

Source: LeRoy and Sonstlie (1981)

- In the US, car ownership steadily reached lower income groups
- In 2017, 92% US households declared having access to private vehicle
- Cycles can come to an end and repeat with *new* transportation mode

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Role of housing age (Brueckner-Rosenthal, 2010)

- Relative housing age changes over time
 - City expands
 - New housing stock is added at the margin
 - Once housing reaches end of lifetime, new stock replaces old
- High-income households have strong preference for housing services
 - High-income households “follow” new housing stock
- Lifecycle of buildings loops over income downtown
 - Suburbanization when housing stock in downtown gets older
 - Gentrification when downtown housing stock is renewed

Evidence (Brueckner-Rosenthal, 2010)

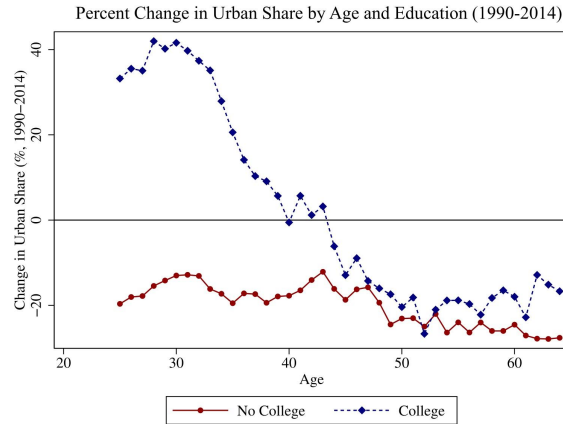
- Housing age is a key factor in explaining income distribution
 - Older housing in city center attracts lower-income households, vs. newer suburban housing attracts wealthier individuals
 - When downtown is replaced or renovated, reverses the pattern
- Predictions: over the next years, housing dynamics will shift income distributions
 - Suburban areas will age, making them less attractive to wealthy
 - Downtown areas will be renewed, attracting richer households
 - Contribute to cycles of downtown gentrification and reducing income gaps between city centers and suburbs
- *Can you think of examples? Would this hold everywhere?*

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Which group drives gentrification?

(Couture and Handbury, 2020)

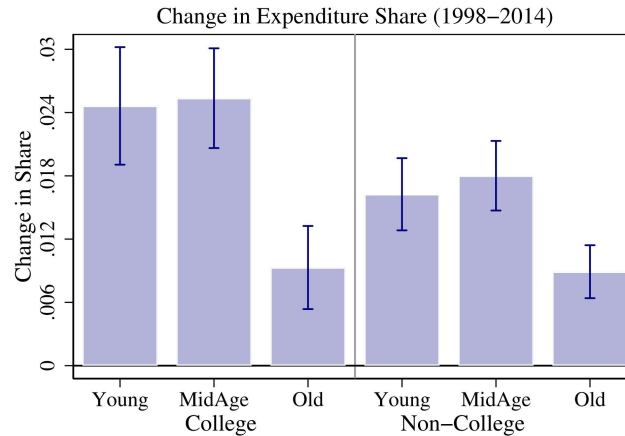


- Downtown gentrification driven by young college-educated individuals
- *What drives these people downtown?*

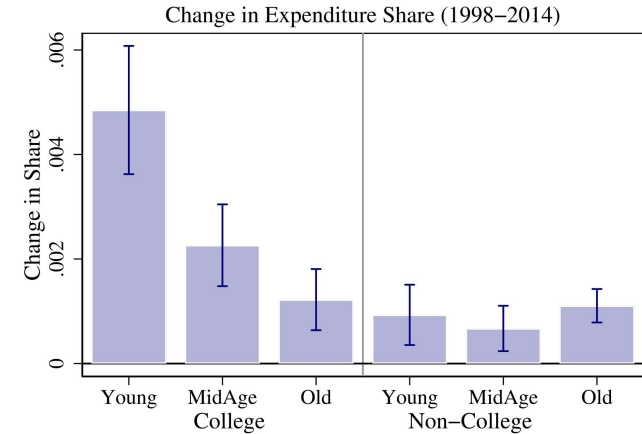
Fig. 2. Notes: The figure shows the percent change between 1990 and 2014 (2012–2016 ACS) of the average urban share of college and non-college educated individuals by age. The data comes from the IPUMS Public Use Microdata Sample and is restricted to the set of 25–64 year-olds in the 27 CBSAs where we can define constant geography urban areas in 1990 and 2012–2016. The urban area of each CBSA is the set of tracts closest to city center that constitute 10% of the total CBSA population in 2000. See [Appendix A](#) for further description of methodology.

Expenditures of young high-skilled (Couture and Handbury, 2020)

Restaurants



Nightlife establishments



- Young high-skilled increasingly spend on urban non-tradable goods

Location of urban amenities (Couture and Handbury, 2020)

Panel A: Initial Levels in Restaurants (dashed) and Food Stores (solid)

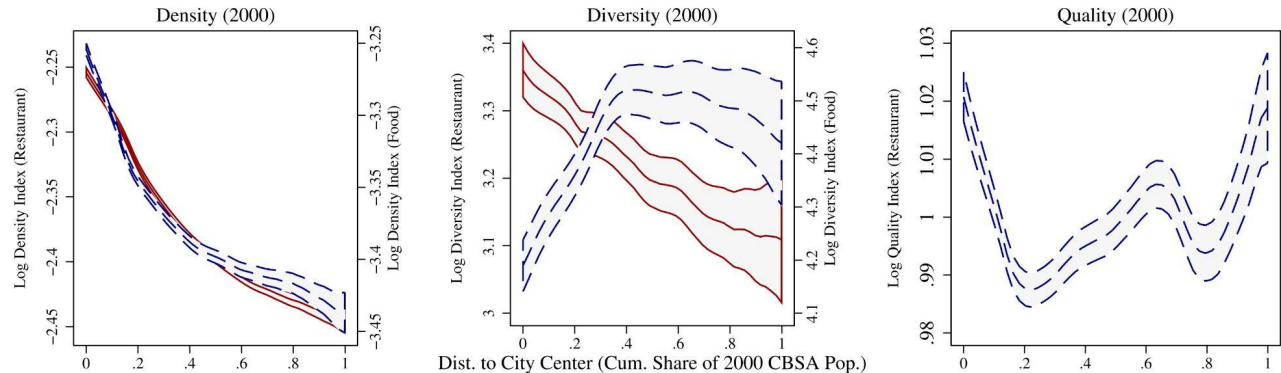


Fig. 4. Notes: Panel A shows a non-parametric kernel fit of the log of amenity density and diversity indexes in 2000 for restaurants and food stores and the amenity quality index in 2000 for restaurants, plotted against the young-college population-weighted distance from the city center for all tracts in our estimation sample. Panel B shows a kernel of the 2000 to 2010 percent change in these indexes. The shaded region around kernel fit depicts the 95% confidence interval. Amenity establishment location is from the NETS data. See [Appendix B](#) for details on consumption amenity index construction.

- Urban non-tradable goods available downtown
- Increasing demand consumption pushes young, high-skilled in centers

The role of urban consumption

- Gentrification and the consumer city
- Young and high-skilled attracted to urban consumption goods
 - Restaurants, museums, theaters and movie theaters,...
 - Endogenous amenities require large consumer base (density)
 - Su (2022): Direct effect of increase in value time for high-skilled amplified by endogenous amenities (residential spillovers)
- Other factors: Expansion of working hours, crime reduction, ...
- Gentrification happens in areas with high amenity value (close to points of interest, historic fabric,...)

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Appreciation during real estate booms

(Guerrieri et al., 2013)

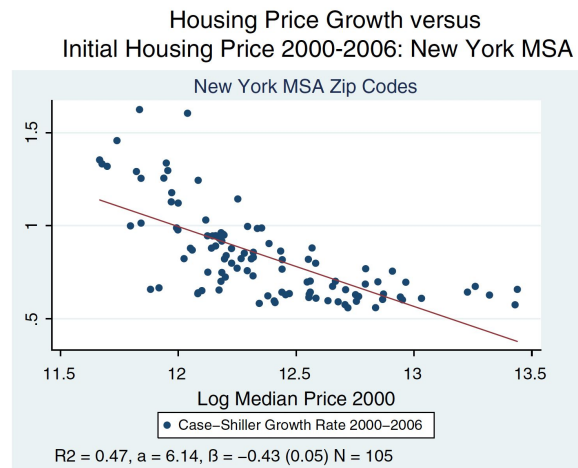


Fig. 1. Figure shows the initial house price in a zip code in 2000 versus the subsequent house price growth in that zip code between 2000 and 2006. The sample includes all zip codes within the New York metro area for which a Case-Shiller house price index exists. We measure the initial house price using median home value from the 2000 Census. We use the Case-Shiller index to compute the growth rate in house price between 2000 and 2006.

- During booms, low house price areas appreciate much faster than average
- Appreciation 2000-2006:
 - NYC: 80%
 - Harlem: 130%
 - Midtown: 45%
- Prime areas less dynamic in booms but more resilient during downturns

Appreciation during real estate booms (Guerrieri et al., 2013)

Table 3

Regression of neighborhood house price appreciation on distance to nearest high-price neighborhood and other controls, across different samples with different house price measures.

Time period	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Log distance to nearest high-price Neighborhood	-0.061 (0.021)	-0.062 (0.019)	-0.044 (0.029)	-0.067 (0.037)	-0.231 (0.042)	-0.136 (0.032)	-0.140 (0.029)
Log distance to nearest high-price Neighborhood*city wide bust indicator	-	-	-	0.070 (0.044)	0.068 (0.051)	0.077 (0.029)	0.083 (0.032)
House price measure/ Neighborhood aggregation	C-S Zip Code	C-S Zip Code	C-S Zip Code	C-S Zip Code	Census Tract	Census Tract	Census Tract
Time period	00-06	00-06	90-00	90-00	90-00	90-00	80-90
Vector of Z controls included	No	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	236	236	223	223	3099	7955	4253
Mean log distance to nearest High-price neighborhood	1.23	1.23	1.22	1.22	0.401	0.499	0.322
Std. dev. log distance to nearest High-price neighborhood	0.524	0.524	0.488	0.488	0.778	0.719	0.716

Note: Table shows regression of neighborhood house price appreciation between period t and $t+k$ on log distance to nearest high price neighborhood within the neighborhood's city, city fixed effects, and a vector of neighborhood controls. High price neighborhoods are those neighborhoods that are within the top quartile of average neighborhood house prices in year t . We restrict our analysis in this table to those neighborhoods within the city which were in the bottom half of the house price distribution in period t . See text for additional sample descriptions and discussion of the controls included. Robust standard errors, clustered by city, are shown in parentheses. All regressions are weighted by the number of owner occupied housing units in the neighborhood in the initial year.

- Places near high-price neighborhoods appreciate faster

Model of endogenous gentrification

(Guerrieri et al., 2013)

- Assumptions:
 - Linear city (each point in the line is a location)
 - Households perfectly mobile
 - Developers build housing at constant marginal cost (constant density)
 - Households differ by income
 - All households prefer to live closer to rich neighbors (*why?*)
 - Poorer value proximity to the rich less (pay lower premium for it)

Model of endogenous gentrification

(Guerrieri et al., 2013)

- Static predictions (spatial equilibrium)
- Rich households live close to each other
- Poor households live further away from rich neighborhoods
- House prices offset neighborhood externality (positive spillovers that rich neighbors impose on others)
 - House prices are highest in rich neighborhoods
 - Decline in distance from rich neighborhoods
 - Fall to the level of marginal housing costs when spillovers fade out

Equilibrium (Guerrieri et al., 2013)

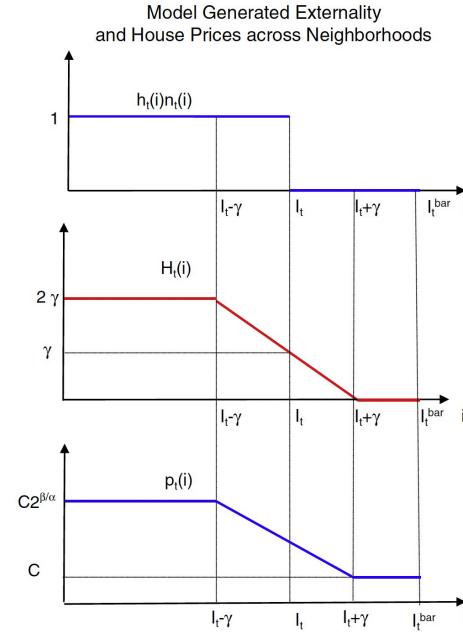


Fig. 2. Figure shows the model predicted relationship between the size of the rich neighborhood (top panel), the value of the neighborhood externality (middle panel), and the house price in the neighborhood (bottom panel). For this figure, we assume $C^r = C^b$. This is done for illustrative purposes.

Equilibrium (Guerrieri et al., 2013)

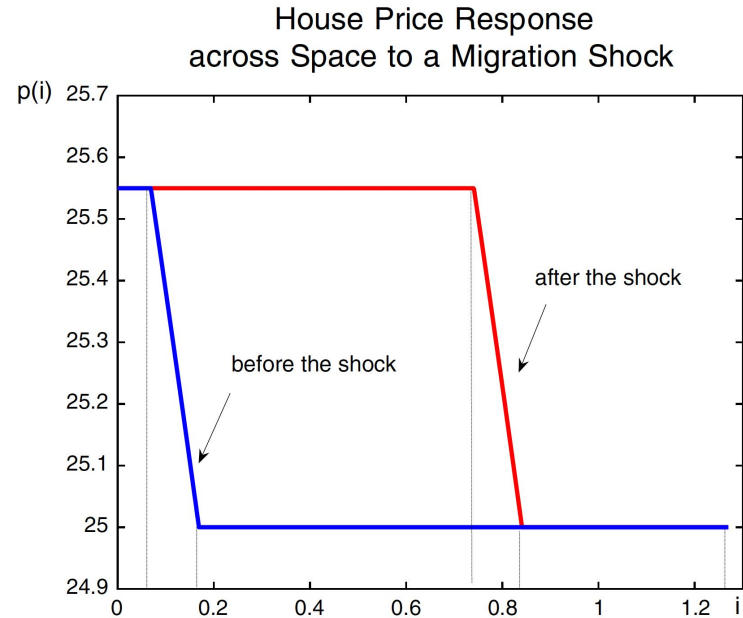


Fig. 3. We set $\alpha = .8$, $\beta = .8$, $\delta^R = .2$, $\delta^P = 0$, $A = 1$, $\gamma = .1$, $r = .03$, $y^R = 1$, $y^P = .5$, $C^R = C^P = 25$, $N^R = N^P = .5$. The shock is an unexpected and permanent increase in ϕ from $\phi = 1$ to $\phi = 5$.

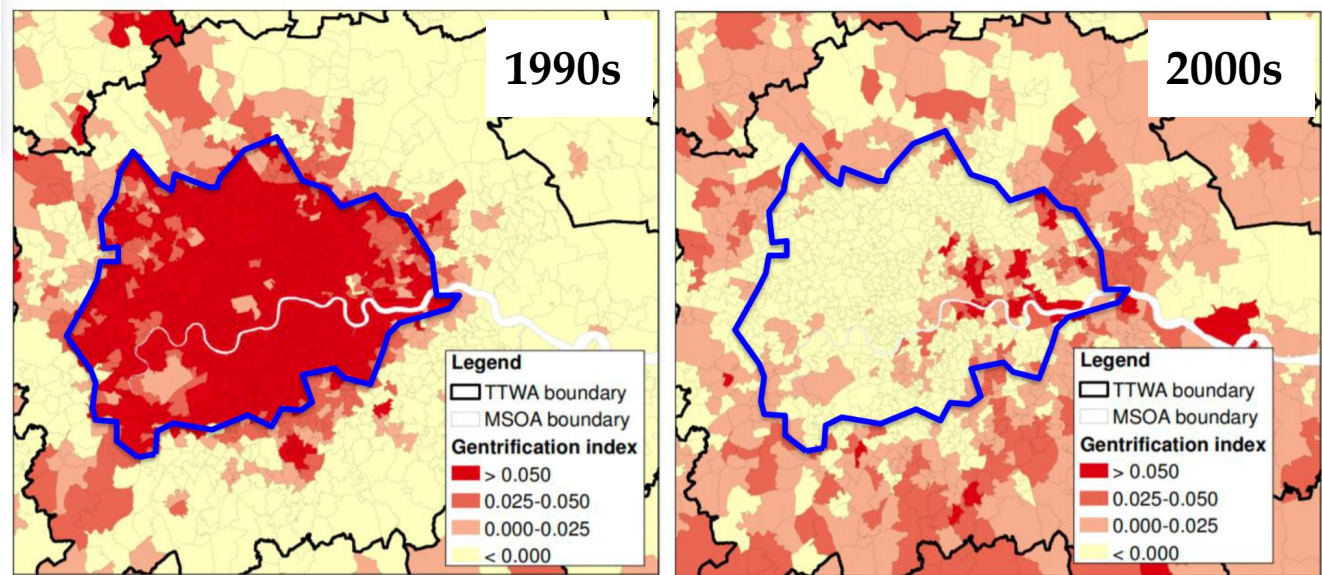
- New rich (e.g., immigrants) locate as close as possible to rich neighborhood
- Pushes rich neighborhood boundary and externality out
- Next generation of “new rich” will locate further out etc

Endogenous gentrification

(Guerrieri et al., 2013)

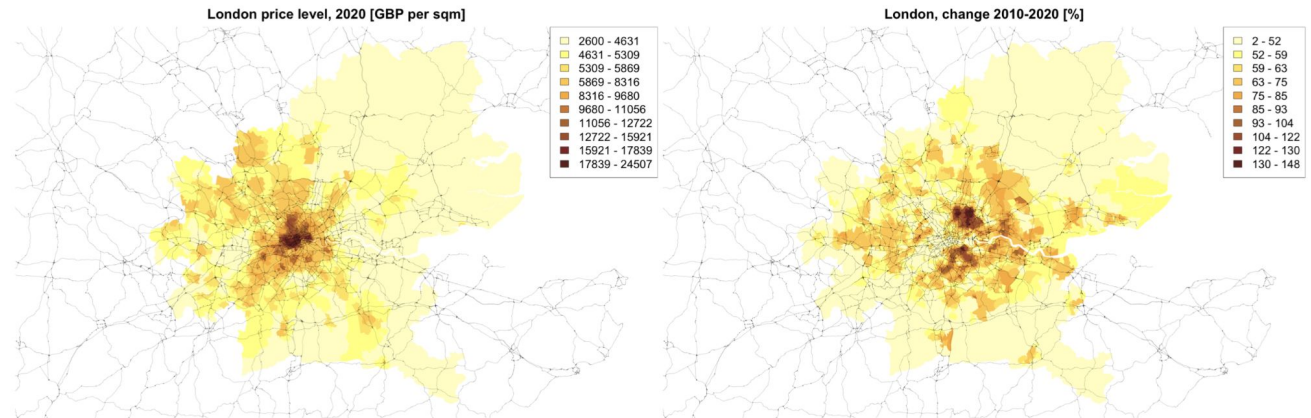
- Guerrieri et al. (2013) develop a model with neighborhood externalities
 - People like to live close to rich neighbors
 - Equilibrium characterized by spatial income segregation
- “New rich” (e.g., immigrants)
 - Move close to rich neighborhood, but not (yet) rich
 - Neighborhood appreciates
 - Poor households displaced to cheaper locations with lower rich-neighborhood spillovers
 - Cycle continues
 - Gentrification is spatially “contagious”

Gentrification index (Waights, 2018)



- Areas gentrified in 1990s were mostly no longer gentrifying in 2000s
- Gentrification moved to adjacent areas

London house prices (Ahlfeldt et al., 2023)



- Gentrification moved to adjacent areas

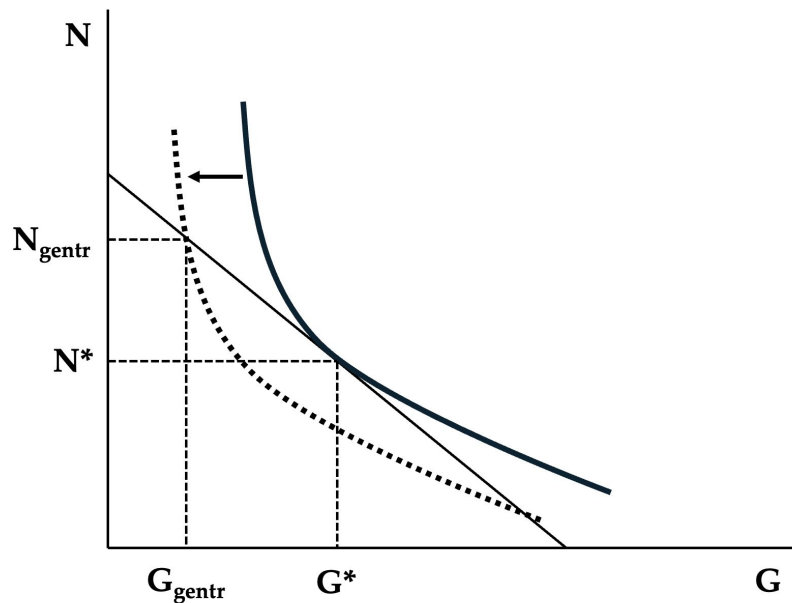
Why does this matter for real estate?

- Correctly predicting gentrification helps generating returns on investment
 - Target areas:
 - To-be-gentrified areas must be below average income areas
 - With some buildings approaching end of their lifecycle
 - With fundamental amenities like historic districts
 - In close proximity to richer areas
 - Ideally, with some urban consumption amenities already there
 - Areas that will benefit from the next transport revolution
 - Marginal areas (gentrifying or close-to) offer greatest potential vs. established ones are safe havens
-

Today: Gentrification and Suburbanization

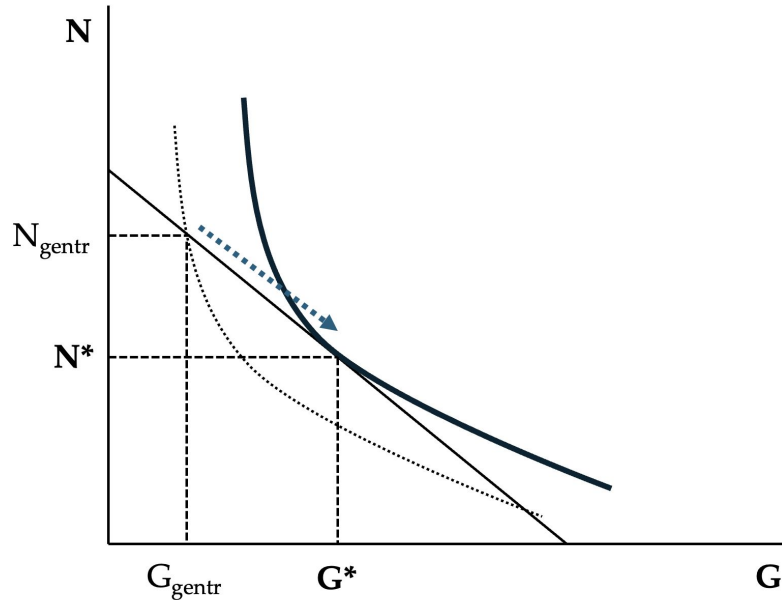
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Gentrification and renter utility



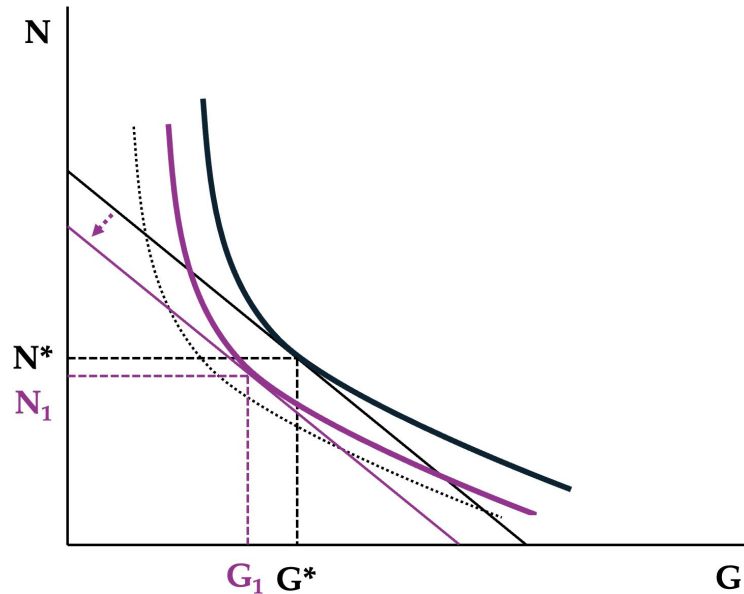
- Consume income on neighborhood quality (N) and non-housing goods (G)
- $Y = nN + G$ ($p_G = 1$)
- Choose nhbd quality that max utility
- $N \uparrow$ (*gentrification*): renter consumes too much of N
- Nhbd quality level N no longer optimal

Gentrification and renter utility



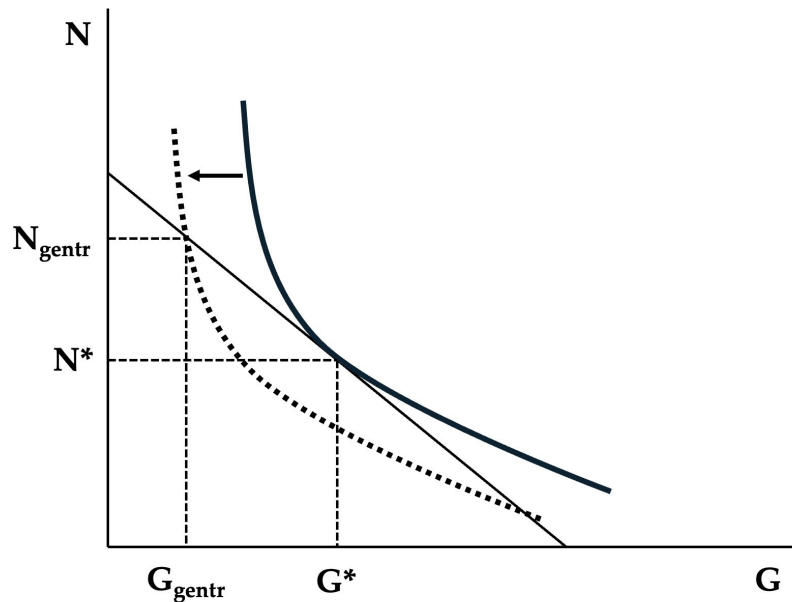
- **Scenario 1: Mobility is costless**
- Renter can move to a different neighborhood with utility-maximizing neighborhood quality N^* to restore utility level

Gentrification and renter utility



- **Scenario 2: Mobility is costly & renter moves**
- Renter moves to a different neighborhood, but the move is costly hence it reduces lifetime budget
- New optimum given $Y \downarrow$: (N_1, G_1) , U lower

Gentrification and renter utility

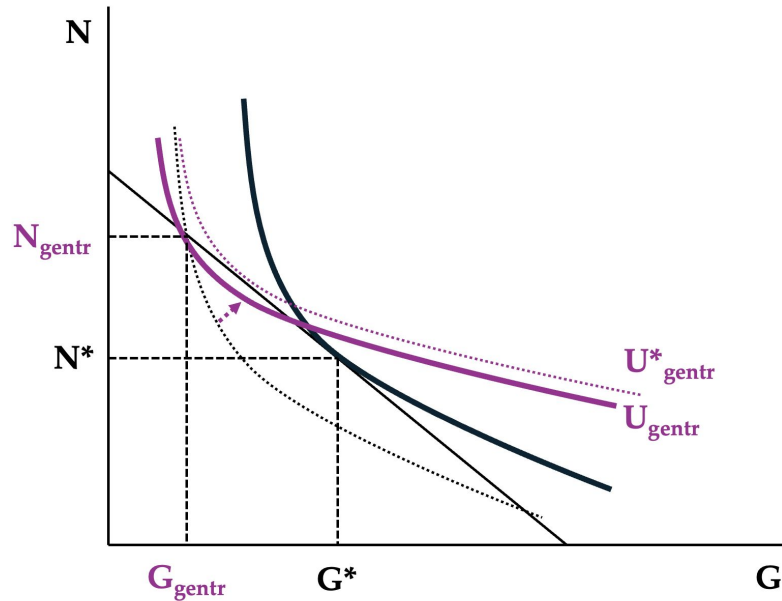


- **Scenario 3: Mobility is costly & renter stays**
- “Forced” to consume neighborhood quality level equal to $N_{\text{gentr}} > N^*$
- Utility is lower than before (given the budget line, utility is max at N^*, G^*)

Gentrification and renter utility

- **Scenario 3a:** Mobility is costly & renter stays + preferences adjust
- *What is likely to happen? How would the previous graph change?*

Gentrification and renter utility

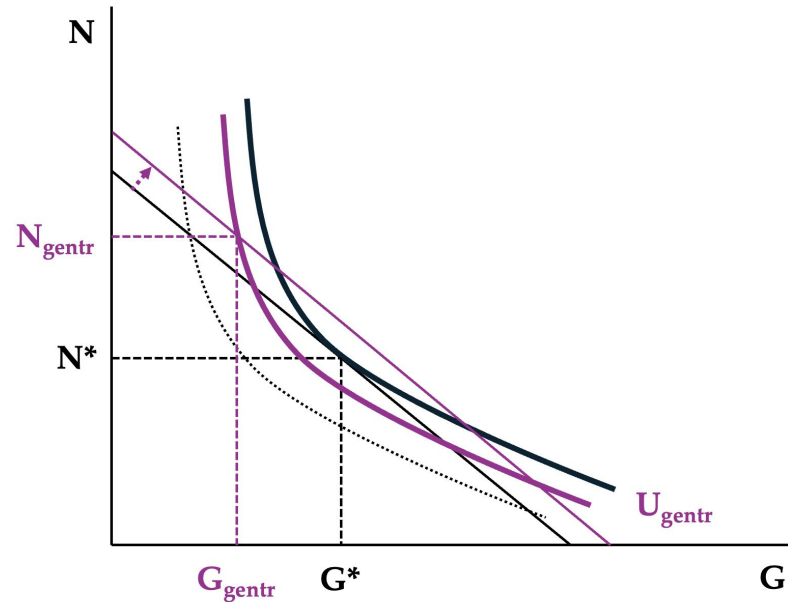


- **Scenario 3a:** Mobility is costly & renter stays + preferences adjust
- Drop in utility relative to the new optimum is less severe than if renter would not have adjusted preferences

Gentrification and renter utility

- **Scenario 3b:** Mobility is costly & renter stays + positive externalities
- *What is likely to happen? How would the previous graph change?*

Gentrification and renter utility

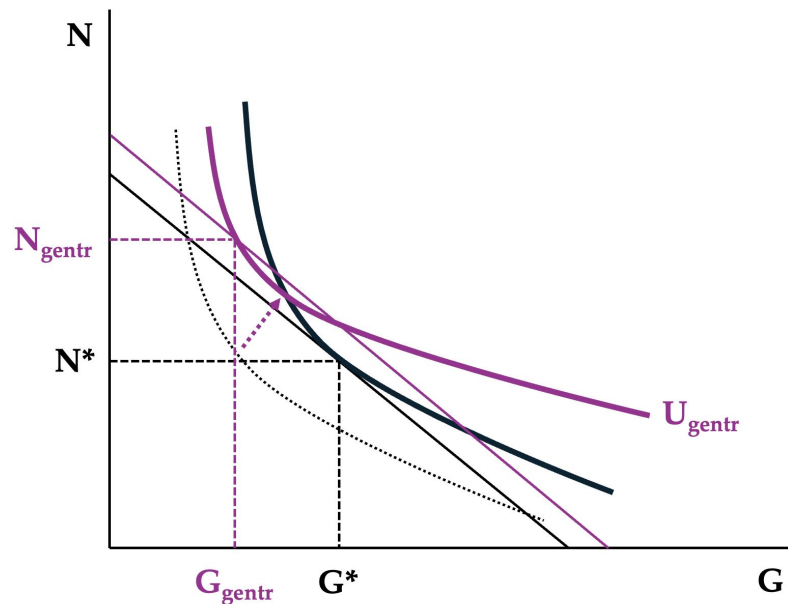


- **Scenario 3b:** Mobility is costly & renter stays + positive externalities
- Renter experiences positive externalities leading to higher lifetime income
- Drop in utility less severe relative to new optimum

Gentrification and renter utility

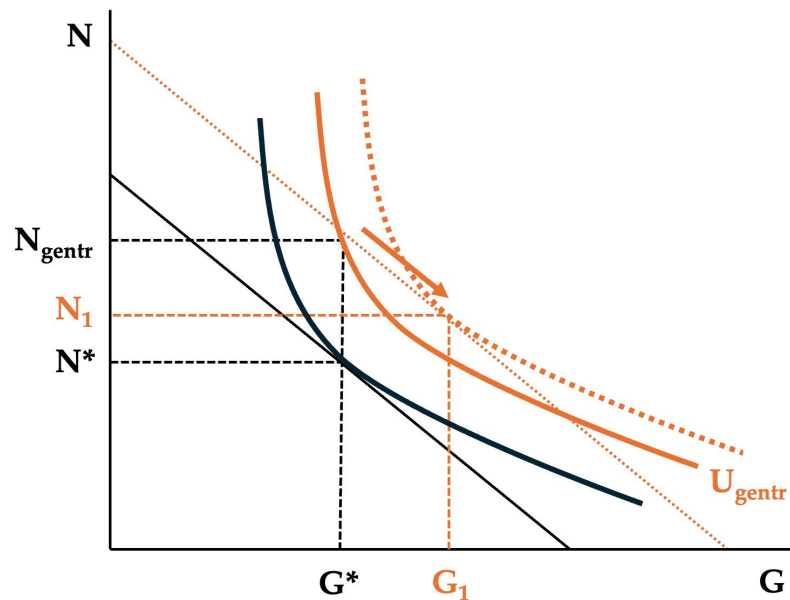
- **Scenario 3c:** Mobility is costly & renter stays + positive externalities + preferences adjust
- *What is likely to happen? How would the previous graph change?*

Gentrification and renter utility



- **Scenario 3c: Mobility** is costly & renter stays + positive externalities + preferences adjust
- Renter experiences positive externalities leading to higher lifetime income and adjusts preferences
- Drop in utility even less severe (under some cases, utility could increase)

Gentrification and homeowner utility



- $N \uparrow$ for free, $U \uparrow$
- Can optimize by moving to a different neighborhood and benefit from $\uparrow$ wealth
- Or, stay: benefit from preference adjustment and/or neighborhood spillovers
- Can incur in cost: loss of social/locational capital (whether stay or move)

Support vs resistance to nhbd changes

- Homeowners should support initiatives that increase property value
- vs. Renters have more reasons to oppose urban renewal policies
- Expected negative effects from gentrification for renters seem to dominate
 - The same positive price signal should be supported by homeowners (homevoters) and opposed by renters (lease voters) and vice versa
 - e.g., evidence from aircraft noise (Ahlfeldt and Maennig, 2015)

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Displacement

- Evidence for the US
 - Gentrification has moderate to no effects on out-migration
 - Neighborhood change driven by in-migration
 - Gentrification reduces white flight (Ellen and O'Regan, 2011)
 - Gentrification of poor Black neighborhoods creates neighborhoods that are attractive to middle class Black households
- Evidence for the UK
 - Gentrification leads to significant displacement of former residents (Waights, 2018)

Effects on stayers

- Stayers experience positive spillovers from gentrification
 - Higher gains in income (Ellen and O'Regan, 2011)
 - Greater neighborhood satisfaction (Ellen and O'Regan, 2011)
 - Better schooling outcomes for children (more likely to attend and complete college)
- Employment grows faster than average in gentrifying neighborhoods
 - More jobs in restaurants/services, less in manufacturing/wholesale
 - Residents lose jobs in nhbd but gain them further away
 - Gentrification reduces crime (Autor et al., 2017)
- Net effect depends on housing costs (+/- for renters)

Summary

- Determinant of dynamic spatial income pattern
 - New expensive transportation mode pushes rich to suburbs
 - Lifecycle of building supply
 - Consumption amenities attract young college-graduate to centers
 - Gentrifiers attracted to rich neighborhoods due to spillover effects
- Welfare effects of gentrification
 - Renters lose unless benefit from spillovers or adjust preferences
 - Homeowners benefit except for loss of social capital
- Evidence points to limited displacement effects and benefits to stayers
 - Stayers may still be net losers if they rent

Next week: Measuring cities with new technologies

- Introduction
 - Improving understanding through measurement
- Detecting urban expansion and land use patterns
- Mapping accessibility and human mobility
- Spatial externalities: Congestion, health, and the environment
- Challenges and the future of cities
 - Implications for the future of real estate

Thank you!

Readings and references

Required readings

- Guerrieri, V., Hartley, D., and Hurst, E. (2013). Endogenous gentrification and housing price dynamics. *Journal of Public Economics*, 100, 45-60.
- LeRoy, S. F., and Sonstelie, J. (1983). Paradise lost and regained: Transportation innovation, income, and residential location. *Journal of Urban Economics*, 13(1), 67-89.

Other readings and references

- Ahlfeldt, G. M., Carozzi, F., and Makovsky, L. (2023). A micro-geographic house price index for England and Wales. CEP WP N. 61.
- Ahlfeldt, G. M., Maennig, W. (2015). Homevoters vs. leasevoters: A spatial analysis of airport effects. *Journal of Urban Economics*, 87, 85-99.
- Autor, D. H., Palmer, C., Pathak, P. (2017). Gentrification and the amenity value of crime reductions: Evidence from rent deregulation. NBER WP 23914.
- Brueckner, J. K., and Rosenthal, S. S. (2009). Gentrification and Neighborhood Housing Cycles: Will America's Future Downtowns be Rich? *The Review of Economics and Statistics*, 91(4), 725-743.
- Couture, V., and Handbury, J. (2020). Urban revival in America. *Journal of Urban Economics*, 119, 103267.

Readings and references

- Cutler, D. M., Glaeser, E. L., and Vigdor, J. L. (1999). The rise and decline of the American ghetto. *Journal of political economy*, 107(3), 455-506.
- Ellen, I. G. and O'Regan, K. M. (2011). How low income neighborhoods change: Entry, exit, and enhancement. *Regional Science and Urban Economics*, 41(2), 89-97.
- Su, Y. (2022). The rising value of time and the origin of urban gentrification. *American Economic Journal: Economic Policy*.
- Waights, S. (2018) Does Gentrification Displace Poor Households? An 'Identification-Via-Interaction' Approach. CEP DP 1540.